Permutations Accelerate Approximate Bayesian Computation

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Abstract

Approximate Bayesian Computation (ABC) methods have become essential tools for performing inference when likelihood functions are intractable or computationally prohibitive. However, their scalability remains a major challenge in hierarchical or high-dimensional models. In this paper, we introduce *permABC*, a new ABC framework designed for settings with both global and local parameters, where observations are grouped into exchangeable compartments. Building upon the Sequential Monte Carlo ABC (ABC-SMC) framework, permABC exploits the exchangeability of compartments through permutation-based matching, significantly improving computational efficiency.

We then develop two further, complementary sequential strategies: *Over-Sampling*, which facilitates early-stage acceptance by temporarily increasing the number of simulated compartments, and *Under-Matching*, which relaxes the acceptance condition by matching only subsets of the data.

These techniques allow for robust and scalable inference even in high-dimensional regimes. We also clarify the connection between permABC and Wasserstein ABC at the level of compartment-level empirical measures, and discuss the asymptotic behaviour of the method as the number of compartments or the over-sampling size grows. Through synthetic and real-world experiments — including a hierarchical Susceptible-Infectious-Recovered model of the early COVID-19 epidemic across 94 French departments — we demonstrate the practical gains in accuracy and efficiency achieved by our approach.

Keywords: Sequential Monte Carlo; Hierarchical models; Permutation-based inference; Likelihood-free methods; Exchangeability

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Introduction

Approximate Bayesian Computation (ABC) (Marin et al. 2012) has emerged as a pivotal tool for statistical inference in scenarios where the likelihood function is computationally infeasible or explicitly unavailable. Initially applied in population genetics by Tavaré et al. (1997) and extended by Pritchard et al. (1999), ABC has found applications across various fields such as climate science (Holden et al. 2018), ecology (Piccioni et al. 2022), and epidemiology (Sisson et al. 2018). The key idea behind ABC is to simulate data under the prior predictive distribution and retain simulations $\mathbf{z}$ that are close to the observed data $\mathbf{y}$, thereby avoiding explicit likelihood computations. This comparison relies on a metric d and a tolerance threshold $\varepsilon > 0$: we accept when $d(\mathbf{y}, \mathbf{z}) \leq \varepsilon$. ABC methods are thus particularly useful for highly complex models, for which it is difficult to calculate the likelihood but easy to simulate synthetic data.

Challenges arise for large datasets and for large parameter space. To reduce the dimensionality of the data, summary statistics are typically employed. Practitioners aim to select informative summaries and to choose ε as small as computationally feasible, as tighter thresholds yield posterior approximations of higher fidelity. The problem of constructing such summaries has received considerable attention. For example, Fearnhead & Prangle (2012) proposed an inferential criterion for summary statistic selection to improve posterior accuracy. Many applications rely on a small number of expert-informed summary statistics.

Significant methodological progress has been made by integrating Monte Carlo techniques into ABC, to facilitate the exploration of the parameter space. Among these, Markov Chain Monte Carlo (MCMC) and Sequential Monte Carlo (SMC) strategies have been particularly influential. Marjoram et al. (2003) proposed ABC-MCMC, which uses a Markov chain to guide parameter proposals toward regions of high posterior mass. Later, Sisson et al. (2007) introduced the Partial Rejection Control (PRC-ABC) algorithm, adopting a particle-based perspective. However, this method suffered from biased weighting issues (Sisson et al. 2009). To resolve these limitations, Beaumont et al. (2009) proposed ABC-PMC, which combines importance sampling and adaptive kernel scaling. This innovation led to improved efficiency and more accurate posterior approximations. Subsequently, Del Moral et al. (2012) provided a theoretical framework for ABC-SMC using backward kernels and adaptive thresholding, which further enhanced robustness and convergence. These algorithmic refinements, particularly in the context of SMC, have made ABC applicable to increasingly complex models and larger datasets. Still, the field continues to evolve, as illustrated by recent work on guided proposals and adaptivity (Picchini & Tamborrino 2024).

Despite these advances, standard ABC approaches remain challenged by high-dimensional problems and parameter dependencies. To address this, Gibbs-inspired strategies have been proposed, which capitalize on model structure to update parameters conditionally. Clarté et al. (2021) and Rodrigues et al. (2020) introduced independently a likelihood-free Gibbs sampling scheme where each parameter block is updated using its approximate conditional distribution via ABC. Clarté et al. (2021) established theoretical convergence guarantees for ABC-Gibbs under partial independence, demonstrating substantial gains in efficiency and approximation quality, particularly in hierarchical contexts. These contributions expand the ABC toolbox toward more scalable and modular inference schemes.

Motivated by these developments, we turn our attention to a specific class of hierarchical models commonly encountered in applied settings where observations are naturally grouped into independent subpopulations or *compartments*. As a motivating example, to which we return in Section 3.5, consider the popular Susceptible-Infectious-Recovered model in epidemiology: we wish to apply this model to the early phases of the COVID epidemic in France, with data observed locally at the level of each of 94 *départements*; initial exploration shows that certain parameters are global (constant across the whole country) but others are local (they vary between the départements). This structure suggests a new inference framework that exploits exchangeability across compartments.

More generally, we consider a setting where the data are divided into K compartments. In each compartment $k = 1, \dots, K$, we observe n data points $\mathbf{y}_k = (y_k^1, \dots, y_k^n) \in \mathcal{D}$, which depend on a shared global parameter β (with prior π_β) and a compartment-specific local parameter μ_k. We assume that $(\mu_k, \mathbf{y}_k)_{k=1}^K$ are exchangeable conditional on β; a typical application is when the local parameters are independent and identically distributed $\mu_1, \dots, \mu_K \sim \pi_\mu$. Collectively, the full dataset is denoted by $\mathbf{y} := (\mathbf{y}_1, \dots, \mathbf{y}_K) \in \mathcal{D}^K$, and the parameter vector by $\boldsymbol{\theta} = (\beta, \mu_1, \dots, \mu_K) \in \Theta$, with total dimension $d = Kd_{\text{loc}} + d_{\text{glob}}$, where d_{loc} and d_{glob} are respectively the dimensions of the local and global parameters. Such a hierarchical model is illustrated in Figure 1.

Let $\mathcal{S}_K$ be the symmetric group over $\{1, \dots, K\}$. We define the permuted parameter vector $\boldsymbol{\theta}_\sigma = (\beta, \mu_{\sigma(1)}, \dots, \mu_{\sigma(K)})$, where $\sigma \in \mathcal{S}_K$ is a permutation of the compartments. We assume that each $\mathbf{y}_k$ is generated independently from a common likelihood function g, conditional on its corresponding parameters. In this context, exchangeability implies that the joint data distribution is invariant under permutations: $f(\mathbf{y} \mid \boldsymbol{\theta}) = \prod_{k=1}^K g(\mathbf{y}_k \mid \beta, \mu_k) = f(\mathbf{y}_\sigma \mid \boldsymbol{\theta}_\sigma)$ for any $\sigma \in \mathcal{S}_K$.

We further assume that the likelihood g is intractable or computationally costly, though we can simulate from it. That is, given (β, μ_k), we can generate pseudo-data $\mathbf{y}_k$. By abuse of notation, we refer to the global simulator as the global likelihood f throughout.

Building on the exchangeability property, the main innovation of our work is to simulate synthetic data, then apply a well-chosen permutation to the compartments before checking the ABC acceptance criterion, yielding our *permABC* algorithm, introduced in Section 1. This strategy increases the acceptance probability by several orders of magnitude. We consider sequential improvements in Section 2: we first introduce permABC-SMC, then

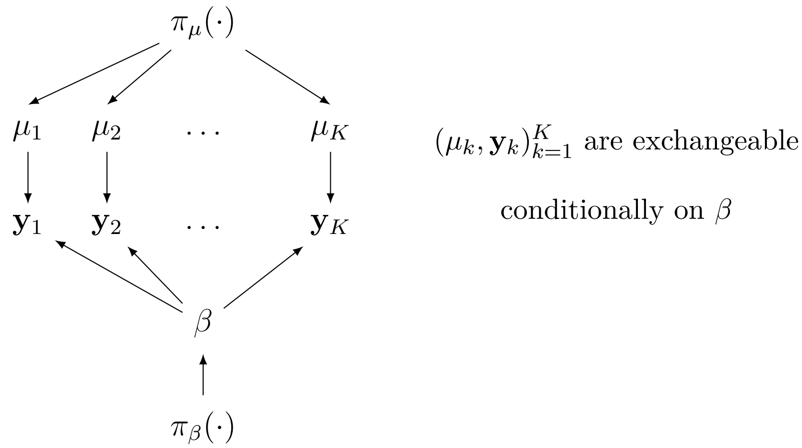


Figure 1: Hierarchical model framework with exchangeable priors: $\mu_k \sim \pi_\mu(\cdot)$, $\beta \sim \pi_\beta(\cdot)$.

define new sequences of intermediate distribution for sequential strategies (Over-Sampling and Under-Matching); they are made possible by our use of permutations, and they improve efficiency and robustness in challenging regimes such as outliers or model mis-specification. Section 3 provides a thorough empirical assessment of the proposed methods, including comparisons with classical ABC techniques, synthetic experiments highlighting the gains of permutation-based inference, and a real-world application to COVID-19 epidemic data in France.

1 From ABC to permABC

In this section, we will review the ABC framework and highlight its limitations within the hierarchical model framework. Subsequently, we will introduce our permutation-based method, permABC, which aims to address these limitations.

1.1 Introduction to ABC

The standard rejection ABC algorithm, often referred to as *Vanilla ABC*, proceeds by drawing parameters from the prior distribution, $\boldsymbol{\theta} \sim \pi(\cdot)$, and then simulating synthetic data from the likelihood, $\mathbf{z} \sim f(\cdot \mid \boldsymbol{\theta})$. The pair $(\boldsymbol{\theta}, \mathbf{z})$ is accepted if the summary statistics of the simulated data are sufficiently close to those of the observed data, that is, if $d(\eta(\mathbf{y}), \eta(\mathbf{z})) \leq \varepsilon$ for some tolerance level $\varepsilon > 0$ and some summary statistic η. This procedure yields samples from the approximate joint distribution $\pi_\varepsilon\{\boldsymbol{\theta}, \mathbf{z} \mid \eta(\mathbf{y})\}$; we marginalize out $\mathbf{z}$ to get the ABC posterior

$$\begin{aligned} \pi_\varepsilon(\boldsymbol{\theta} \mid \eta(\mathbf{y})) &\propto \int_{\mathcal{D}^K} \pi_\varepsilon(\boldsymbol{\theta}, \mathbf{z} \mid \eta(\mathbf{y}))\, d\mathbf{z} \\ &\propto \pi(\boldsymbol{\theta}) \int_{A_{\mathbf{y},\varepsilon}} f(\mathbf{z} \mid \boldsymbol{\theta})\, d\mathbf{z}, \end{aligned} \tag{1}$$

where $A_{\mathbf{y},\varepsilon} = \{\mathbf{z} \in \mathcal{D}^K \mid d(\eta(\mathbf{y}), \eta(\mathbf{z})) \leq \varepsilon\}$ denotes the acceptance region.

Combining an informative summary statistic η with a small tolerance ε typically leads to a good approximation of the true posterior distribution, so that $\pi_\varepsilon(\boldsymbol{\theta} \mid \eta(\mathbf{y})) \approx \pi(\boldsymbol{\theta} \mid \mathbf{y})$. This approximate posterior—often called a *pseudo-posterior*—is obtained by replacing the true likelihood $f(\mathbf{y} \mid \boldsymbol{\theta})$ with the integral of the likelihood over the neighborhood $A_{\mathbf{y},\varepsilon}$ of the observed data, effectively using it as a surrogate for the intractable likelihood.

Throughout this work, we assume for clarity that the data are compared directly, i.e., $\eta = \mathrm{Id}$. Under this assumption, the acceptance region simplifies to the ε-ball $B_\varepsilon(\mathbf{y})$ centered at $\mathbf{y}$ with respect to the distance d. Nonetheless, the proposed method naturally extends to the use of low-dimensional summary statistics.

In our numerical examples, we take $\mathcal{D} = \mathbb{R}^n$ and use the distance $d(\cdot,\cdot)$ between two datasets in $\mathcal{D}^K$, each consisting of $K \times n$ observations, defined as follows:

$$d(\mathbf{y}, \mathbf{z}) = \sqrt{\sum_{k=1}^{K} w_k^2 \|\mathbf{y}_k - \mathbf{z}_k\|_2^2}, \tag{2}$$

where $\|\cdot\|_2$ denotes the Euclidean norm, and the weights w_k are used to control the relative influence of each compartment in the distance computation. The weight vector $w_{1:K}$ can either be manually specified by the user or selected adaptively to prevent any single compartment from dominating the distance, as suggested by Prangle (2017).

This naive ABC algorithm faces several limitations, including poor scalability in the parameter space and an exponentially increasing computational cost due to the declining acceptance rate as ε decreases. In the context of the hierarchical model of Figure 1, the acceptance rate further diminishes as the number of compartments K increases. Simulating a dataset $\mathbf{z}$ from the prior predictive distribution that matches all compartments of $\mathbf{y}$ within a small tolerance ε can be particularly challenging.

1.2 Motivations of permABC

A common approach in ABC to increase the acceptance rate is to exploit the exchangeability of the compartments through a permutation-invariant summary statistic, such as the sorted vector. For instance, when there is a one-dimensional observation (or statistic) per compartment, $\mathbf{y} \in \mathbb{R}^K$, it is natural to use the ordering $\eta(\mathbf{y}) = \mathrm{sort}(\mathbf{y})$ as a summary. This replaces the $K!$ possible labelings by a canonical representation, thereby simplifying the comparison between datasets.

Moreover, in the univariate case, sorting is known to minimize the distance over all permutations of the simulated data. Specifically, $d\big(\mathrm{sort}(\mathbf{y}), \mathrm{sort}(\mathbf{z})\big) = \min_{\sigma \in \mathcal{S}_K} d(\mathbf{y}, \mathbf{z}_\sigma)$, where $\mathcal{S}_K$ denotes the set of all permutations of $\{1, \dots, K\}$. Equivalently, in this one-dimensional setting, comparing sorted vectors amounts to computing the Wasserstein distance between the empirical measures of the compartments, which connects our construction to Wasserstein ABC (WABC) (Bernton et al. 2019).

However, this equivalence should not obscure an important difference in inferential target. WABC is designed to compare samples viewed as unordered point clouds, and is therefore primarily suited to settings where inference concerns global features of the data distribution. By contrast, our setting is hierarchical: inference concerns not only the global parameter β, but also the local parameters $\mu_{1:K}$, at least up to relabelling. In particular, WABC is not designed for hierarchical models with compartment-specific local parameters, since it does not keep track of which simulated compartment is associated with which local parameter.

In higher dimension, the scalar sorting argument no longer applies. When each $\mathbf{y}_k$ is a vector in $\mathbb{R}^n$ with $n > 1$, there is no canonical ordering that is both deterministic and permutation-optimal. We therefore solve the matching problem directly, which leads to the acceptance criterion

$$\min_{\sigma \in \mathcal{S}_K} d(\mathbf{y}, \mathbf{z}_\sigma) \leq \varepsilon. \tag{3}$$

This condition means that the simulated dataset is accepted whenever it matches the observed dataset up to a permutation of the compartments. It can therefore yield a substantial increase in acceptance probability, heuristically of order $K!$ when ε is small and the neighborhoods associated with distinct permutations are essentially disjoint. The optimization problem in (3) can be formulated as a linear sum assignment problem, or equivalently a bipartite matching problem, and solved efficiently using a modified Jonker–Volgenant algorithm (Jonker & Volgenant 1987), with computational complexity $\mathcal{O}(K^3)$ (Crouse 2016). Although this remains substantial, it reduces the combinatorial cost from $K!$ to polynomial time and makes problems with up to $K = 100$ compartments tractable in practice; see the real-data application in Section 3.5.

This matching criterion is closely related to optimal transport between the empirical distributions of the compartments. In the multivariate case, Bernton et al. (2019) discuss several cheaper approximations to exact transport, including Hilbert-curve sorting, swap-based refinements, and entropically regularized Sinkhorn divergences. These approximations are particularly attractive when the number of points is very large. In our applications, however, K remains only moderately large, while the dimension of each compartment-level observation may be non-negligible. In this regime, solving the assignment problem exactly remains practical and robust. We nevertheless return to this point in Section 2, where we show that within an SMC scheme a cheaper swap-based refinement, warm-started from the previous iteration's optimal permutation, is sufficient in practice.

However, using (3) does not yield samples from the pseudo-posterior defined in (1), but rather from the following permutation-invariant version:

$$\begin{aligned}\tilde{\pi}_\varepsilon(\boldsymbol{\theta} \mid \mathbf{y}) &\propto \int_{\mathcal{D}^K} \tilde{\pi}_\varepsilon(\boldsymbol{\theta}, \mathbf{z} \mid \mathbf{y})\, d\mathbf{z} \\ &\propto \pi(\boldsymbol{\theta}) \int_{\tilde{A}_{\mathbf{y},\varepsilon}} f(\mathbf{z} \mid \boldsymbol{\theta})\, d\mathbf{z},\end{aligned} \tag{4}$$

where the acceptance region is

$$\tilde{A}_{\mathbf{y},\varepsilon} = \left\{ \mathbf{z} \in \mathcal{D}^K \;\middle|\; \min_{\sigma \in \mathcal{S}_K} d(\mathbf{y}, \mathbf{z}_\sigma) \le \varepsilon \right\} = \bigcup_{\sigma \in \mathcal{S}_K} B_\varepsilon(\mathbf{y}_\sigma),$$

with $B_\varepsilon(\mathbf{y}_\sigma)$ denoting the ε-ball centered at $\mathbf{y}_\sigma$ for the distance d.

In this formulation, the likelihood $f(\mathbf{y} \mid \boldsymbol{\theta})$ is approximated by its integral over a neighborhood of all permutations of the observed dataset:

$$\int_{\tilde{A}_{\mathbf{y},\varepsilon}} f(\mathbf{z} \mid \boldsymbol{\theta})\, d\mathbf{z} \approx \sum_{\sigma \in \mathcal{S}_K} f(\mathbf{y}_\sigma \mid \boldsymbol{\theta}).$$

Inference under (4) is therefore naturally defined only up to a permutation of the local components $\mu_{1:K}$, as in WABC (Bernton et al. 2019), effectively yielding posterior samples of the form $\tilde{\boldsymbol{\theta}} = (\beta, \{\mu_1, \ldots, \mu_K\})$, where the local parameters are treated as an unordered set. This target may be appropriate when the local parameters are nuisance quantities, but in many hierarchical applications one wishes to recover inference on the ordered local components as well.

To our knowledge, this specific permutation-based formulation of ABC for hierarchical models with local parameters has not been formally studied in the literature. We therefore introduce *permABC*, which exploits permutation matching to improve acceptance while making it possible to recover inference on the full parameter vector $\boldsymbol{\theta}$, including the ordered local parameters.

1.3 The permABC algorithm

Restoring identifiability among compartments is essential for enabling inference on the full parameter vector. One natural way to achieve this is by applying a suitable permutation to the accepted parameters, such that the simulated data are optimally aligned with the observed dataset. However, explicitly testing all $K!$ permutations is computationally intractable.

Fortunately, whenever the acceptance condition (3) is satisfied, there exists at least one permutation of the simulated dataset that meets the ABC criterion, namely the optimal one minimizing the distance. This permutation, denoted by $\sigma^* = \arg\min_{\sigma \in \mathcal{S}_K} d(\mathbf{y}, \mathbf{z}_\sigma)$, provides a natural way to reorder the parameter vector.

In practice, the *permABC* algorithm proceeds similarly to standard ABC: we sample $\boldsymbol{\theta} \sim \pi(\cdot)$ and simulate data $\mathbf{z} \sim f(\cdot \mid \boldsymbol{\theta})$. If the minimum distance over all permutations satisfies the tolerance condition, we accept and return the pair $(\boldsymbol{\theta}_{\sigma^*}, \mathbf{z}_{\sigma^*})$. This induces a new pseudo-posterior distribution, denoted $\pi_\varepsilon^*(\cdot, \cdot \mid \mathbf{y})$, which corresponds to the pushforward of the distribution $\tilde{\pi}_\varepsilon$ defined in (4) under the optimal alignment map $\varphi_{\mathbf{y}}$:

$$\pi_\varepsilon^*(\cdot, \cdot \mid \mathbf{y}) := \varphi_{\mathbf{y}\sharp} \tilde{\pi}_\varepsilon(\cdot, \cdot \mid \mathbf{y}) \tag{5}$$

where

$$\varphi_{\mathbf{y}} : (\boldsymbol{\theta}, \mathbf{z}) \mapsto (\boldsymbol{\theta}_{\sigma^*}, \mathbf{z}_{\sigma^*}).$$

By construction, $\varphi_{\mathbf{y}} \circ \varphi_{\mathbf{y}} = \varphi_{\mathbf{y}}$, so $\varphi_{\mathbf{y}}$ acts as a projection. Moreover, its preimage satisfies $\varphi_{\mathbf{y}}^{-1}(\{\boldsymbol{\theta}, \mathbf{z}\}) = \{(\boldsymbol{\theta}_\sigma, \mathbf{z}_\sigma) \mid \sigma \in \mathcal{S}_K\}$.

As a result, the projected pseudo-posterior satisfies:

$$\begin{aligned}\pi_\varepsilon^*(\boldsymbol{\theta}, \mathbf{z} \mid \mathbf{y}) &\propto \tilde{\pi}_\varepsilon(\varphi_{\mathbf{y}}^{-1}(\{(\boldsymbol{\theta}, \mathbf{z})\}) \mid \mathbf{y}) \cdot \mathbb{I}_{\mathrm{Im}(\varphi_{\mathbf{y}})}(\boldsymbol{\theta}, \mathbf{z}) \\ &\propto \sum_{\sigma} \tilde{\pi}_\varepsilon(\boldsymbol{\theta}_\sigma, \mathbf{z}_\sigma \mid \mathbf{y}) \cdot \mathbb{I}_{\mathrm{Im}(\varphi_{\mathbf{y}})}(\boldsymbol{\theta}, \mathbf{z}) \\ &\propto \tilde{\pi}_\varepsilon(\boldsymbol{\theta}, \mathbf{z} \mid \mathbf{y}) \cdot \mathbb{I}_{\mathrm{Im}(\varphi_{\mathbf{y}})}(\boldsymbol{\theta}, \mathbf{z}).\end{aligned}$$

This restricts accepted samples to the image of $\varphi_{\mathbf{y}}$. We define the constrained space as

$$\mathcal{D}_{\mathbf{y}}^{*K} = \left\{ \mathbf{z} \in \mathcal{D}^K \mid d(\mathbf{y}, \mathbf{z}) = \min_{\sigma \in \mathcal{S}_K} d(\mathbf{y}, \mathbf{z}_\sigma) \right\}.$$

Consequently, the marginal pseudo-posterior can be expressed as:

$$\pi_\varepsilon^*(\boldsymbol{\theta} \mid \mathbf{y}) \propto \pi(\boldsymbol{\theta}) \int_{A^*_{\mathbf{y},\varepsilon}} f(\mathbf{z} \mid \boldsymbol{\theta})\, d\mathbf{z}, \tag{6}$$

where the new acceptance region is

$$A^*_{\mathbf{y},\varepsilon} = B_\varepsilon(\mathbf{y}) \cap \mathcal{D}_{\mathbf{y}}^{*K}.$$

The *permABC* algorithm (see Algorithm 1) therefore defines a different approximation of the posterior distribution than classical ABC methods. In particular, it yields a different posterior approximation $\pi_\varepsilon^*(\cdot \mid \mathbf{y})$, which we shall compare to the vanilla ABC approximation $\pi_\varepsilon(\cdot \mid \mathbf{y})$.

1.4 Comparison with ABC and asymptotic perspective

The permutation step in permABC has two distinct asymptotic interpretations, depending on whether one focuses on the global parameter β or on the labelled local parameters $\mu_{1:K}$.

For the global parameter β, once the labels of the local components are ignored, the acceptance rule of permABC falls within the same Wasserstein-type paradigm as WABC (Bernton et al. 2019). In our hierarchical setting, the number of compartments K plays the role of the sample size in WABC, since the comparison is performed through the empirical distributions of the observed and simulated compartments. This viewpoint is useful because

Algorithm 1. permABC Vanilla

Input: observed dataset $\mathbf{y}$, number of iterations N, threshold $\varepsilon > 0$.

Output: a sample $(\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(N)})$ from the pseudo-posterior distribution $\pi_\varepsilon^*(\cdot \mid \mathbf{y})$.

for $i = 1, \ldots, N$ **do**

  repeat

    $\boldsymbol{\theta} \sim \pi(\cdot)$;

    $\mathbf{z} \sim f(\cdot \mid \boldsymbol{\theta})$;

    $\sigma^* = \arg\min_{\sigma \in \mathcal{S}_K} d(\mathbf{y}, \mathbf{z}_\sigma)$;

  until $d(\mathbf{y}, \mathbf{z}_{\sigma^*}) \leq \varepsilon$;

  $\boldsymbol{\theta}^{(i)} = \boldsymbol{\theta}'_{\sigma^*}$

it allows us to directly invoke the asymptotic intuition developed for WABC: when $\varepsilon \to 0$ with the observations kept fixed, the pseudo-posterior approaches the exact posterior, while as the sample size increases one obtains posterior concentration under suitable identifiability conditions. In our context, this supports the view that the permutation-invariant marginal pseudo-posterior in β inherits the same qualitative behaviour as K increases. Moreover, when ε is kept fixed, the resulting target should be interpreted as a coarsened posterior rather than as a Dirac mass, in the spirit of Miller & Dunson (2019).

For the labelled local parameters $\mu_{1:K}$, however, the relevant comparison is no longer with WABC but with standard ABC. Indeed, permABC differs from standard ABC in one key aspect: instead of retaining all permutations that satisfy the ABC criterion, it keeps only the optimal one, namely the permutation that minimizes the discrepancy defined in Equation (3). This choice is computationally attractive, but it alters the pseudo-posterior by constraining the pseudo-data to lie in $\mathcal{D}_{\mathbf{y}}^{*K}$. A stratified variant recovering the standard pseudo-posterior is described in Appendix **??**. We now show that, for small enough ε, this difference vanishes.

Lemma 1.1. *Let $\mathbf{y} \in \mathcal{D}^K$ denote the observed dataset, and define*

$$\varepsilon^* := \frac{1}{2} \min_{\sigma \neq \mathrm{Id}} d(\mathbf{y}, \mathbf{y}_\sigma).$$

Then, for any $\varepsilon < \varepsilon^*$ and any simulated dataset $\mathbf{z} \in \mathbb{R}^{K \times n}$, at most one permutation satisfies the ABC criterion:

$$\mathrm{Card}\,\{\sigma \in \mathcal{S}_K : d(\mathbf{y}, \mathbf{z}_\sigma) \leq \varepsilon\} \leq 1.$$

The proofs of the preceding lemma, the following theorem, and the subsequent corollary are deferred to Appendix **??**.

We assume that no two compartments have exactly the same observed data. This implies that $\varepsilon^* > 0$, and this condition is satisfied almost surely for continuous data. Under this assumption, permABC and standard ABC coincide for sufficiently small ε, as stated below.

Theorem 1.2. *Let $\pi^*_\varepsilon(\cdot \mid \mathbf{y})$ denote the pseudo-posterior distribution resulting from permABC, and $\pi_\varepsilon(\cdot \mid \mathbf{y})$ that obtained from standard ABC. Then, for all $\varepsilon < \varepsilon^*$,*

$$\pi^*_\varepsilon(\cdot \mid \mathbf{y}) = \pi_\varepsilon(\cdot \mid \mathbf{y}).$$

This equivalence ensures that permABC inherits the same small-tolerance asymptotic behaviour as standard ABC for the full labelled parameter vector. In particular, any consistency result for ABC immediately transfers to permABC in the regime $\varepsilon \to 0$.

Corollary 1.3. *If the pseudo-posterior distribution $\pi_\varepsilon(\cdot \mid \mathbf{y})$ obtained by standard ABC converges to the true posterior distribution as $\varepsilon \to 0$, then the pseudo-posterior distribution $\pi^*_\varepsilon(\cdot \mid \mathbf{y})$ obtained by permABC also converges to the true posterior distribution as $\varepsilon \to 0$.*

Thus, the efficiency gains provided by permABC come at no cost to asymptotic consistency. More generally, beyond mere consistency, one may appeal to the asymptotic theory developed for ABC by Frazier et al. (2018), which gives general results on posterior concentration, limiting posterior shape, and asymptotic distributions under suitable regularity, identifiability, and joint (n, ε)-asymptotics. In the present hierarchical setting, these results apply to permABC in the regime $\varepsilon < \varepsilon^*$, where permABC and ABC coincide, while the WABC viewpoint described above provides the appropriate asymptotic interpretation for the marginal pseudo-posterior in β.

2 Sequential strategy with permutations

We present three complementary strategies for accelerating ABC via permutations within an SMC framework. Section 2.2 embeds the permABC criterion of Section 1 into the adaptive ABC-SMC algorithm of Del Moral et al. (2012). Section 2.4 introduces *Over-Sampling*, which simulates $M > K$ components per particle and selects the K best-matching ones, while Section 2.5 presents *Under-Matching*, which begins with $L < K$ components and progressively increases L toward K. Both extensions exploit the permutation structure to improve the acceptance rate at each SMC iteration.

2.1 Sequential ABC: state of the art and limitations

Rejection-based ABC methods remain computationally expensive due to their reliance on the prior as proposal. ABC-SMC algorithms (Sisson et al. 2007, Beaumont et al. 2009, Del Moral et al. 2012) address this by constructing a sequence of intermediate pseudo-posteriors $\pi_{\varepsilon_0}, \ldots, \pi_{\varepsilon_T}$ with decreasing thresholds, combined with importance sampling and MCMC moves. We adopt the adaptive framework of Del Moral et al. (2012) with $M = 1$ dataset per proposal (Bornn et al. 2017), yielding binary weights. Since the ESS then equals the number of surviving particles and may not reflect true sample diversity, we also monitor the *unique particle rate*—the proportion of distinct parameter values among accepted particles—as a complementary diagnostic.

In the following subsections, we propose three ways to accelerate ABC-SMC with permutations, based on three different strategies to define intermediate distributions.

2.2 The permABC-SMC algorithm

The adaptive SMC algorithm proposed by Del Moral et al. (2012) can be seamlessly extended to the permABC framework. By substituting the standard ABC acceptance criterion with the permutation-based criterion (3), we develop a sequential algorithm that targets the pseudo-posterior distribution $\tilde{\pi}_\varepsilon(\boldsymbol{\theta} \mid \mathbf{y})$.

Upon reaching the stopping criterion or when $\varepsilon_T = \varepsilon$, we apply the projection step (5) to each particle in the final population $\{(\boldsymbol{\theta}^{T,i}, \mathbf{z}^{T,i})\}_{i=1}^N$. This yields samples from the permuted pseudo-posterior $\pi^*_\varepsilon(\boldsymbol{\theta} \mid \mathbf{y})$, effectively restoring identifiability among compartments.

It is essential to ensure that the proposal distributions $q_t(\cdot)$ are adapted to the new acceptance criterion. We detail such kernels in Section 2.6, emphasizing their role in maintaining the efficiency of the sampling process under the permutation-based framework.

This approach, termed *permABC-SMC*, facilitates a sequential transition from the prior to the permuted pseudo-posterior $\pi^*_\varepsilon(\boldsymbol{\theta} \mid \mathbf{y})$. Thanks to proposal kernels tailored to the permutation-based selection rule, permABC-SMC enhances the exploration of the parameter space, particularly in high-dimensional settings where standard ABC methods may struggle.

The permABC-SMC approach is a natural way to extend permABC into a sequential method. It relies on intermediate distributions of the form $(\pi^*_{\varepsilon_t})_t$ which are similar to Over-Sequential ABC methods, and it already leads to substantial computational gains, as shown in Section 3. Before we examine these practical results, we introduce two innovative sequences of intermediate distributions, with novel selection criteria which are only available because we rely on permutation-based matching.

2.3 Fast assignment via warm-started swaps

Solving the exact linear sum assignment problem (3) costs $\mathcal{O}(K^3)$ per particle at every MCMC step, which can dominate the overall runtime in sequential schemes. However, within an SMC iteration the particle configuration varies slowly: the random-walk kernel perturbs the local parameters by a small amount and the tolerance ε_t decreases gradually. As a consequence, the optimal permutation σ^* of a proposed particle is typically close to the permutation already associated with its current state.

This observation motivates a warm-started refinement strategy: at each proposed move, we initialise the assignment with the current σ^* and apply a few sweeps of pairwise swaps, each sweep running in $\mathcal{O}(K^2)$. Swaps are accepted greedily whenever they decrease the permuted distance, following the construction of Bernton et al. (2019) but warm-started from the previous optimum rather than from a Hilbert ordering. In practice, one or two sweeps are sufficient to recover σ^* or a permutation inducing the same acceptance decision, so that an exact LSA call is only required at the first SMC iteration or when large jumps occur. This reduces the amortised per-step assignment cost from $\mathcal{O}(K^3)$ to $\mathcal{O}(K^2)$ without altering the target pseudo-posterior.

2.4 Over-Sampling

The Over-Sampling strategy leverages the flexibility of the permABC framework to sample more efficiently from the target pseudo-posterior $\pi^*_\varepsilon(\cdot \mid \mathbf{y})$ by simulating more compartments than are present in the observed data. Specifically, instead of generating K pseudo-compartments, we simulate $M > K$ compartments: that is, we consider $\mathbf{z} \in \mathcal{D}^M$ while the observed data remain in $\mathcal{D}^K$. We then accept or reject based on the best simulated subset of size K.

To implement this strategy, we draw $\boldsymbol{\theta} = (\beta, \mu_1, \ldots, \mu_M) \sim \pi(\cdot)$, simulate the corresponding data $\mathbf{z} = (z_1, \ldots, z_M) \sim f(\cdot \mid \boldsymbol{\theta})$, and use the following acceptance condition:

$$\min_{\sigma \in \mathcal{A}_K^M} d(\mathbf{y}, \mathbf{z}_\sigma) \leq \varepsilon, \tag{7}$$

where $\mathcal{A}_K^M$ denotes the set of injections from $\{1, \ldots, K\}$ to $\{1, \ldots, M\}$, i.e., the set of partial K-permutations of $\{1, \ldots, M\}$. As in (3), this optimization problem can be solved efficiently using the same assignment algorithm with computational complexity $\mathcal{O}(K^2 M)$ (Crouse 2016).

We slightly abuse notation by denoting f the joint simulator over compartments, regardless of whether the number of compartments is K (observed) or M (simulated). This is justified since the simulator is defined componentwise via a common mechanism g, applied independently to each compartment: formally, $f(\mathbf{z} \mid \boldsymbol{\theta}) := \prod_{m=1}^{M} g(z_m \mid \beta, \mu_m)$.

Analogously to the case $M = K$, this procedure defines a pseudo-posterior distribution $\tilde{\pi}_\varepsilon^M(\cdot \mid \mathbf{y})$, and its projected version $\pi_\varepsilon^{*M}(\cdot \mid \mathbf{y})$, which is obtained by applying the optimal partial permutation minimizing (7) in a reasoning analogous to that of Section 1 (see Appendix **??** for details).

More precisely, Over-Sampling can be viewed as a pushforward construction. Let $\sigma^* = \sigma^*(\boldsymbol{\theta}, \mathbf{z}, \mathbf{y}) \in \arg\min_{\sigma \in \mathcal{A}_K^M} d(\mathbf{y}, \mathbf{z}_\sigma)$, using any measurable tie-breaking rule, and define the matching map

$$\Psi_{\mathbf{y}}^{(M)} : (\boldsymbol{\theta}, \mathbf{z}) \longmapsto (\boldsymbol{\theta}^*, \mathbf{z}^*), \tag{8}$$

where

$$\mathbf{z}^* = \mathbf{z}_{\sigma^*} \in \mathcal{D}^K, \qquad \boldsymbol{\theta}^* = \Big(\beta, \mu_{\sigma^*(1)}, \ldots, \mu_{\sigma^*(K)}\Big). \tag{9}$$

Then the projected Over-Sampling distribution is exactly the pushforward

$$\pi_\varepsilon^{*M}(\cdot \mid \mathbf{y}) = \Big(\Psi_{\mathbf{y}}^{(M)}\Big)_{\#} \tilde{\pi}_\varepsilon^M(\cdot \mid \mathbf{y}). \tag{10}$$

In other words, $\tilde{\pi}_\varepsilon^M$ is the ABC law on the enlarged space $(\beta, \mu_{1:M}, \mathbf{z}_{1:M})$, while π_ε^{*M} is obtained by projecting onto the K matched compartments and their associated local parameters.

This pushforward representation also yields an asymptotic description of the regime $M \to \infty$. Let $g_\beta(z) := \int g(z \mid \beta, \mu)\, \pi(d\mu)$, $S_\beta := \operatorname{supp}(g_\beta)$, and define the support-based discrepancy $\delta_\beta(\mathbf{y}) := \inf_{x_{1:K} \in S_\beta^K} d(\mathbf{y}, x_{1:K})$. Under mild regularity assumptions (see Appendix **??** for the full derivation), the marginal Over-Sampling pseudo-posterior on the global parameter satisfies

$$\tilde{\pi}_\varepsilon^M(\beta \mid \mathbf{y}) \xrightarrow[M \to \infty]{} \pi(\beta)\, \mathbf{1}\{\delta_\beta(\mathbf{y}) \leq \varepsilon\}\,. \tag{11}$$

In particular, if $\delta_\beta(\mathbf{y}) = 0$ for all β in the prior support, then $\tilde{\pi}_\varepsilon^M(\beta \mid \mathbf{y}) \to \pi(\beta)$: infinite Over-Sampling destroys all information on the global parameter. By contrast, the pushforward map $\Psi_{\mathbf{y}}^{(M)}$ retains the K matched local parameters, which become concentrated as M grows. This regime does not commute with the $K \to \infty$ asymptotics of Section **??**: Over-Sampling is best viewed as an algorithmic bridge rather than an asymptotic regime to push. Figure 2 illustrates both effects on the Gaussian toy model.

This new distribution family is particularly appealing, as it introduces a natural sequence of auxiliary distributions that converge to the target distribution $\pi_\varepsilon^*(\cdot \mid \mathbf{y})$. We define a decreasing sequence $M_0 > M_1 > \cdots > M_T = K$, keeping ε fixed, such that the acceptance condition (7) becomes increasingly selective until it recovers the standard permABC criterion (3) when $M = K$. To navigate this sequence $\left\{\tilde{\pi}_\varepsilon^{M_t}, t = 0, \ldots, T\right\}$, we adapt the ABC-SMC framework of Del Moral et al. (2012), as discussed in Section 2.1. At each stage, we apply the optimal partial permutation σ^* to each accepted particle to obtain a sample from $\pi_\varepsilon^*(\cdot \mid \mathbf{y})$. This results in a new algorithm we call *permABC-Over-Sampling (permABC-OS)*.

This method converges in fewer iterations but raises two practical challenges: (i) calibrating M_0 and ε jointly to ensure a large enough initial population, and (ii) mitigating particle extinction when M_t decreases, since previously matched compartments may no longer be available. We address both through a duplication mechanism and adaptive heuristics for choosing (M_t); details are in Appendix **??**.

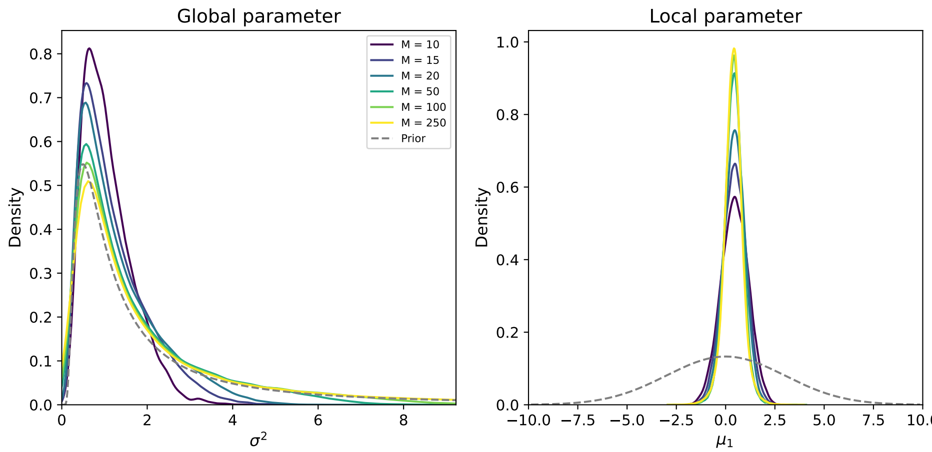


Figure 2: Over-Sampling pushforward marginals for the Gaussian toy model ($K = 10$). Each curve corresponds to one $M \in \{10, \ldots, 250\}$. Left: marginal on σ^2 flattens towards the prior as M grows, in agreement with (11). Right: marginal on μ_1 concentrates as the optimal permutation selects the best match among M candidates.

2.5 Under-Matching

The under matching strategy—referred to as *permABC-Under-Matching (permABC-UM)*—adopts a perspective opposite to Over-Sampling. In this approach, we simulate synthetic data with the same number K of compartments as the observed data; but at intermediate steps, we no longer require the simulated data to match all K observed compartments. Instead, we simulate K compartments but only require $L < K$ to match. The resulting acceptance criterion is defined as:

$$\min_{\sigma \in \mathcal{A}_L^K, \tilde{\sigma} \in \mathcal{A}_L^K} d(\mathbf{y}_{\tilde{\sigma}}, \mathbf{z}_\sigma) \leq \varepsilon, \tag{12}$$

where σ and $\tilde{\sigma}$ denote the sets of L selected compartments from the simulated and observed datasets, respectively.

To solve this optimization problem efficiently, we extend the cost matrix by adding $K - L$ dummy rows and $K - L$ dummy columns with zero cost. These artificial entries ensure that $K - L$ assignments are trivially matched, leaving the algorithm to find the best L true matches. This reduces the partial matching problem to a standard linear sum assignment problem on a square matrix of size $(2K - L) \times (2K - L)$, which can be solved in $\mathcal{O}\{(2K - L)^3\}$ time using classical algorithms (Crouse 2016).

This formulation allows for the algorithm to search over all possible L-sized subsets in both datasets, thereby relaxing the matching constraint and increasing flexibility in early iterations.

Although the distance d in Equation (2) is defined for full datasets in $\mathcal{D}^K$, it naturally extends to subsets. In this context, given $\sigma, \tilde{\sigma} \in \mathcal{A}_L^K$, the restricted distance is such that

$$d(\mathbf{y}_{\tilde{\sigma}}, \mathbf{z}_{\sigma})^2 = \sum_{k=1}^{L} w_{\tilde{\sigma}(k)}^2 \|\mathbf{y}_{\tilde{\sigma}(k)} - \mathbf{z}_{\sigma(k)}\|_2^2.$$

This formulation implies that compartments with higher weights have a greater impact on the acceptance condition, and are thus harder to match. This property can be exploited to prioritize compartments deemed more informative or robust, by appropriately adjusting the weights w_k.

As in other variants of permABC, we employ a SMC scheme through a sequence of intermediate distributions $\tilde{\pi}_{\varepsilon}^{L_t}$, corresponding to an increasing schedule $L_0 < L_1 < \cdots < L_T = K$. This gradual tightening of the matching criterion provides a smooth path toward the final target $\pi_{\varepsilon}^*(\cdot \mid \mathbf{y})$, where all compartments are required to match.

As with Over-Sampling, calibrating ε relative to L_0 is critical: if the model is misspecified for even one compartment, the final transition $L_{T-1} = K - 1 \rightarrow L_T = K$ can cause severe particle degeneracy. Practical guidelines for choosing (L_t) are in Appendix **??**; the benefits of Under-Matching in the presence of outliers are demonstrated in Section 3.4.

2.6 Proposal Kernels for Permuted Inference

We recall that, by the abuse of notation introduced previously, we write $\boldsymbol{\theta}_\sigma = (\beta, \mu_{\sigma(1)}, \ldots, \mu_{\sigma(K)})$ to denote the parameter vector permuted by a given $\sigma \in \mathcal{S}_K$, acting on the local components.

In the ABC-SMC framework of Del Moral et al. (2012), the MCMC kernel K_t at iteration t uses a Gaussian random walk proposal q_t of the form:

$$q_t(\cdot \mid \boldsymbol{\theta}) = \mathcal{N}\Big\{\boldsymbol{\theta}, \operatorname{diag}\Big(\tau_t^2\Big)\Big\},$$

where $\tau_t^2 \in \mathbb{R}^d$ is a vector of componentwise variances. These are typically defined as:

$$\tau_t^2 = 2 \cdot \operatorname{var}\Big\{\Big(\boldsymbol{\theta}^{t-1,i}\Big)_i\Big\},$$

i.e., twice the empirical variance across particles from the previous population. A Metropolis–Hastings correction step is then applied using the ABC acceptance criterion.

In the case of permABC-SMC, the exchangeability of the local parameters introduces label switching in the particle populations, which complicates the direct computation of componentwise variances. To resolve this, we first align the local parameters across particles using their optimal permutations σ_i^*, and compute:

$$\tau_t^2 = 2 \cdot \operatorname{var}\Big\{\Big(\boldsymbol{\theta}_{\sigma_i^*}^{t-1,i}\Big)_i\Big\}.$$

This alignment restores parameter consistency across compartments, and allows us to maintain a diagonal Gaussian kernel with meaningful componentwise variances. The proposal kernel is then given by:

$$q_t(\cdot \mid \boldsymbol{\theta}, \sigma) = \mathcal{N}\Big[\boldsymbol{\theta}, \operatorname{diag}\Big\{\sigma^{-1}(\tau_t^2)\Big\}\Big],$$

where $\sigma^{-1}(\tau_t^2)$ permutes the variance vector accordingly, so that each coordinate of $\boldsymbol{\theta}$ is perturbed using the variance estimated in the appropriate reference frame.

For Over-Sampling and Under-Matching, the same alignment principle applies: matched compartments reuse the aligned variance τ_t^2, while unmatched ones receive a default value $\tau_0^2 := \max_{k \in \mathcal{I}} \tau_t^2[k]$. In the Under-Matching case, the partial permutation $\tilde{\sigma}_i^{-1} \circ \sigma_i$ recovers the correspondence between observed and simulated compartments for each particle. Full details are in Appendix ??.

3 Numerical experiments

3.1 Comparison between ABC and permABC

As discussed in Section 1.4, permABC exhibits a different behavior from standard ABC. In particular, two regimes emerge depending on the value of the threshold ε: when $\varepsilon > \varepsilon^*$, permABC yields a different approximation of the posterior compared to ABC; whereas when $\varepsilon \leq \varepsilon^*$, permABC coincides with standard ABC and thus inherits its convergence guarantees.

To better illustrate the differences between the two approaches, we consider a simple toy model with no global parameter, defined as follows:

$$\mu_k \sim \mathcal{U}(-2, 2), \quad \mathbf{y}_k \sim \mathcal{U}(\mu_k - 1, \mu_k + 1), \quad k = 1, \dots, K,$$

with $K = 2$, and a fixed observation $y = (-1, 1)^\top$.

Figure 3 highlights the two different regimes. When $\varepsilon > \varepsilon^*$, permABC is concentrated in a region constrained by the order statistics, leading to a pseudo-posterior different from that of standard ABC. However, when the threshold satisfies $\varepsilon \leq \varepsilon^*$, both methods recover the same pseudo-posterior. In this example, the critical value is $\varepsilon^* = \sqrt{1/8}/2 \approx 0.177$.

Note that, even in its vanilla form, permABC offers a significant gain in simulation efficiency. Since it effectively avoids redundant permutations during rejection, it reduces the required number of simulations by a factor of approximately $K!$. In this simple example with $K = 2$, we achieve the same approximation quality (i.e., same $\varepsilon < \varepsilon^*$) using only half the number of simulations compared to standard ABC.

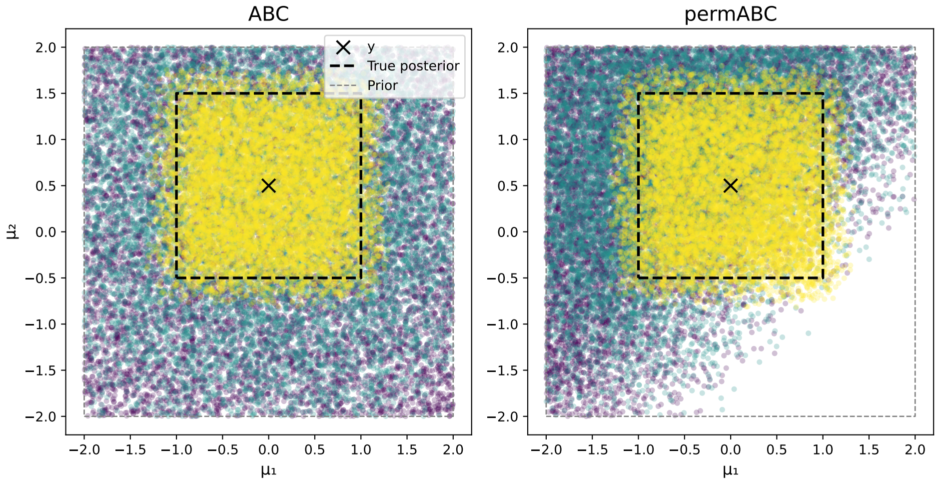


Figure 3: ABC (left) vs. permABC (right) on the uniform toy model with $K = 2$. Dots are sampled (μ_1, μ_2) at three tolerance levels: $\varepsilon = \infty$ (purple), $\varepsilon > \varepsilon^*$ (green), $\varepsilon = \varepsilon^*$ (yellow). Both methods agree when $\varepsilon < \varepsilon^*$; permABC is more concentrated and efficient at equal budget.

3.2 Challenges of High-Dimensional ABC: A Gaussian Toy Model

We now consider a classical hierarchical model commonly used in the Bayesian literature:

$$\beta \sim \mathrm{IG}(a, b), \quad \mu_k \sim \mathcal{N}(0, s^2), \quad \mathbf{y}_k = (y_{1,k}, \ldots, y_{n,k}) \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_k, \beta), \quad k = 1, \ldots, K.$$

Throughout this section we set $a = b = 5$, $s = 10$, $n = 10$ observations per component, and $K = 20$ unless stated otherwise.

This simple yet representative model provides a convenient testbed to compare the various methods discussed in this paper, and to highlight the limitations of standard ABC approaches in high-dimensional settings. All ABC methods involve a fundamental trade-off between the number of simulations N, the tolerance threshold ε, and the effective number of accepted samples (or the number of unique particles in sequential schemes). A more efficient method achieves a lower value of ε for a given simulation budget N and fixed final sample size.

To enable fair comparison across methods, we adopt a consistent evaluation criterion: for each experiment, we report the value of ε achieved when the algorithm yields exactly 1,000 unique particles. In this setting, the number of simulations required to obtain a single unique accepted sample serves as a proxy for the effective sample size (ESS). This pseudo-ESS provides a normalized measure of sampling efficiency, which we use throughout our benchmarks.

A low tolerance ε, however, is only a proxy for posterior quality: two methods that target different pseudo-posteriors cannot be compared on ε alone. To evaluate the quality of the produced approximation directly, we instead report the sliced Wasserstein-2 distance (Bonneel et al. 2015, Nadjahi, Durmus, Chizat, Kolouri, Shahrampour & Şimşekli 2020) between the weighted particle cloud and a reference posterior obtained by long MCMC (PyMC) on the Gaussian toy model of Section 3.2. The sliced Wasserstein metric is well suited to this setting: it metrises weak convergence on $\mathbb{R}^d$, admits linear-time Monte Carlo estimators in the ambient dimension, and has been used as a target discrepancy in ABC by Nadjahi, De Bortoli, Durmus, Badeau & Şimşekli (2020). Reporting SW_2 assesses whether the permutation-based gains in tolerance translate into genuinely better posterior approximations.

Rejection-based algorithms such as vanilla ABC, as well as sequential variants like ABC-SMC and ABC-PMC, are known to scale poorly with the dimension of the parameter space, here indexed by the number of groups K. As K increases, the acceptance rate drops sharply, resulting in a dramatic rise in computational cost. In contrast, the permutation-based acceptance rule introduced in permABC significantly improves the efficiency of both rejection and sequential samplers in high-dimensional regimes. The symmetry structure across local parameters $\mu_1, \ldots, \mu_K$ allows permABC to reduce redundant simulations and produce sharper approximations of the target pseudo-posterior.

Figure 4 illustrates this phenomenon in terms of the sliced Wasserstein-2 distance to the reference posterior: permutation-based methods reach substantially lower SW_2 values than their non-permuted counterparts at equal simulation budget. In particular, for $K = 20$ and without outliers, permABC-SMC and permABC-Vanilla plateau roughly 3–6 times closer to the reference than ABC-SMC or ABC-Vanilla. These gains become increasingly pronounced as K grows, confirming the advantage of permutation-based strategies in hierarchical or exchangeable models.

A complementary comparison in terms of wall-clock time — rather than simulation count — yields similar conclusions, and is provided in Appendix ??.

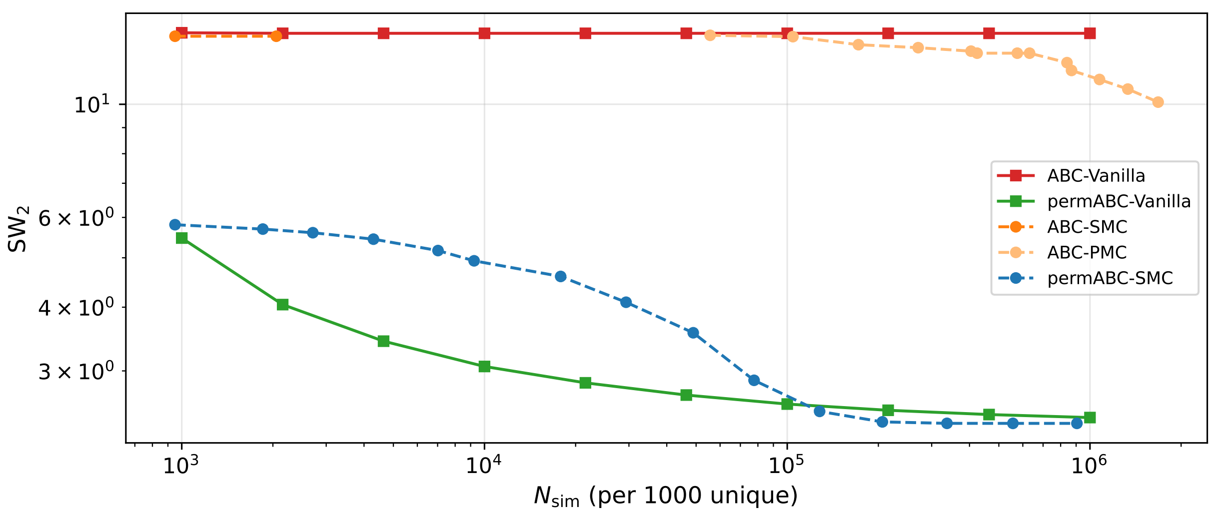


Figure 4: Sliced Wasserstein-2 distance to the reference posterior vs. number of simulations (log–log) for the Gaussian toy model with $K = 20$, no outliers ($N_{\text{particles}} = 5{,}000$). Permutation-based methods reach substantially lower SW_2 at equal budget. Wall-clock time counterpart in Appendix ??.

3.3 Failure cases of ABC-Gibbs

ABC-Gibbs methods (Clarté et al. 2021, Rodrigues et al. 2020) have been designed to handle hierarchical models similar to those we consider. They are however less robust to model dependencies. We illustrate this with a modified hierarchical model designed to highlight the limitations of ABC-Gibbs:

$$\beta \sim \mathcal{N}(0, 10^2), \quad \mu_k \sim \mathcal{N}(0, 10^2), \quad \mathbf{y}_k = (\mathbf{y}_{1,k}, \ldots, \mathbf{y}_{n,k}) \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_k + \beta, 1), \quad k = 1, \ldots, K.$$

This model is deliberately over-parameterized: it includes $K + 1$ parameters for only K observations. Despite its simplicity, the posterior distribution exhibits a strong ridge structure due to the symmetry between local parameters μ_k and the global parameter β. This posterior can be reliably approximated using reference MCMC software such as PyMC.

Figure 5 shows the two-dimensional marginal posterior over (μ_1, β) obtained under a fixed simulation budget. While permABC-SMC provides a conservative and coherent approximation of the target posterior—as expected for an ABC method with global proposals—ABC-Gibbs produces a truncated approximation, failing to explore the full posterior support.

This failure is structural: ABC-Gibbs updates each parameter conditionally on the others, so strong dependencies cause poor mixing. In contrast, permABC jointly considers the full parameter configuration and is robust to such coupling.

3.4 Discussion of the Over-Sampling and Under-Matching Approaches

The Over-Sampling and Under-Matching strategies provide two complementary extensions of the permABC framework, each improving robustness and efficiency in different regimes of hierarchical inference under exchangeability.

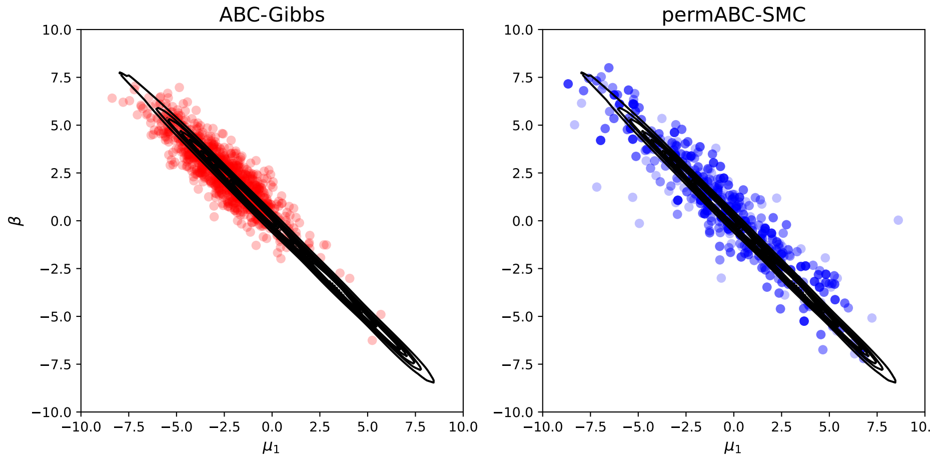


Figure 5: Marginal posterior over (μ_1, β) for the over-parameterised model (3.3). Ellipses: exact MCMC level sets. permABC-SMC (circles) captures the full ridge; ABC-Gibbs (triangles) is trapped in a narrow subset, illustrating poor mixing under strong global–local coupling.

We evaluate their performance on a Gaussian toy model with $K = 20$ compartments, among which 4 have local means μ_k lying in the tails of the prior. This setup simulates a realistic scenario involving mild model misspecification or contamination by outliers. Figure 6 reports the sliced Wasserstein-2 distance SW_2 between the weighted particle cloud and a reference posterior obtained by long MCMC, as a function of the wall-clock time required to obtain 1,000 unique accepted particles. This posterior-quality metric directly measures how close each algorithm gets to the true posterior.

The Over-Sampling method (permABC-OS) reaches values of SW_2 comparable to standard permABC-SMC at the same runtime, and remains the method of choice in the low-tolerance regime: even though each iteration is more expensive because of larger assignment problems and duplicated particles (see Appendix **??**), the additional candidate compartments allow faster progression in the contaminated setting.

Under-Matching (permABC-UM) is the method that benefits most from the presence of outliers. By relaxing the matching constraint to only $L < K$ compartments, it allows rapid convergence in SW_2 in the intermediate-tolerance regime while ignoring the worst-matching ones, in line with the behaviour expected from the asymptotic analysis of Section 2.4. Empirically, the SW_2 advantage of permABC-UM over permABC-SMC grows almost log-linearly with the number of outliers: on this benchmark, a linear fit on log(ratio) gives a slope of roughly -0.12 to -0.16 per additional outlier for $K \in \{10, 15, 20\}$. The same model without contamination does not exhibit such an improvement, confirming that the Under-Matching gain is specifically attributable to outliers. On the other hand, as the algorithm transitions to full matching ($L_t \to K$), the acceptance condition tightens and the particle population can degenerate; unlike in the Over-Sampling strategy, this issue cannot be alleviated by duplication, and the method typically fails to reach the lowest SW_2 values.

These contrasting behaviors suggest that combining the two strategies could yield additional benefits. One possibility is to begin inference with permABC-UM to quickly eliminate poorly fitting regions, then switch to permABC-SMC to refine toward smaller tolerances. More generally, hybrid schemes where L_t and M_t are jointly adapted — potentially alternating with changes in ε — offer a promising direction for future exploration.

A complementary figure reporting the number of simulations (instead of runtime) is provided in Appendix **??**.

3.5 Application to Real Data: The SIR Model

In this final section, we illustrate the applicability of our methodology on real-world data by analysing the early spread of COVID-19 in France using publicly available hospitalization records from Santé publique France (Santé publique France 2020). The dataset provides daily hospital admissions at several geographic levels. We focus on mainland France (excluding overseas territories and Corsica), which yields 94 departmental time series.

To describe the epidemic dynamics, we use a classical Susceptible–Infectious–Recovered

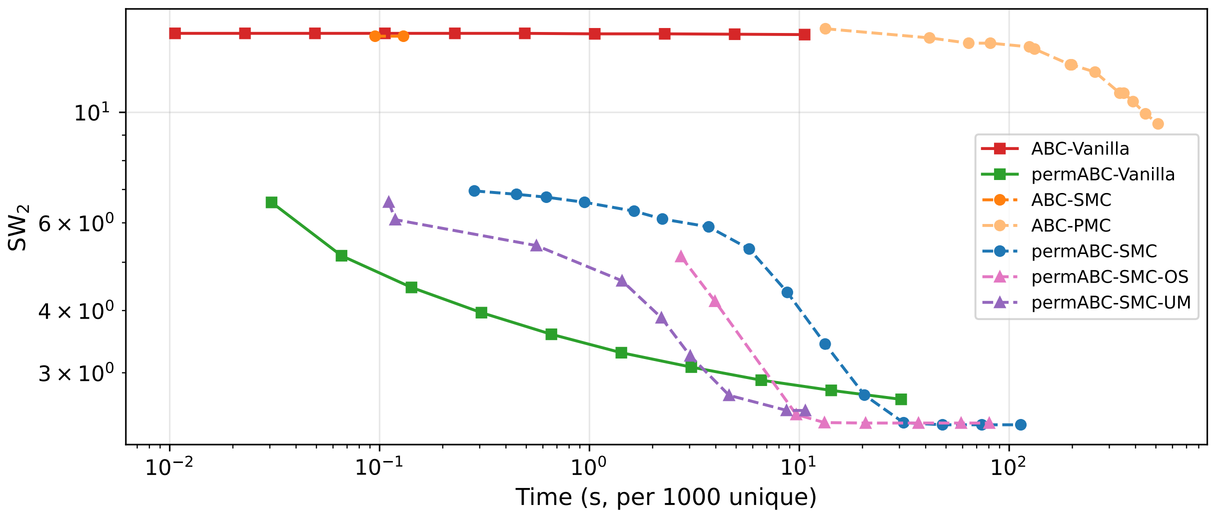


Figure 6: SW_2 to reference posterior vs. wall-clock time (log–log) for the Gaussian toy model with $K = 20$ and 4 outliers ($N_{\text{particles}} = 5{,}000$). Under-Matching is most efficient at intermediate tolerances; Over-Sampling remains competitive at low tolerances. Simulation-count counterpart in Appendix **??**.

(SIR) model, defined through the latent system of ordinary differential equations

$$\begin{aligned} \frac{dS}{dt} &= -\delta \frac{SI}{N}, \\ \frac{dI}{dt} &= \delta \frac{SI}{N} - \nu I, \\ \frac{dR}{dt} &= \nu I, \end{aligned} \tag{13}$$

where S, I, and R denote the numbers of susceptible, infectious, and recovered individuals, respectively, N is the population size, and δ and ν are the transmission and recovery rates. The corresponding basic reproduction number is $R_0 = \delta/\nu$.

Since the data are hospital admissions, the ODE defines a latent trajectory. To introduce stochasticity, the transmission rate is perturbed at each sub-daily step by multiplicative log-normal noise, $\delta_k^{(t)} = \delta_k \exp(\sigma\, \xi_k^{(t)})$, $\xi_k^{(t)} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, with $\sigma = 0.05$. This captures day-to-day variability (reporting delays, behavioural changes, contact patterns), so that the simulated data entering the ABC distance are stochastic.

We restrict attention to the first wave (March–July 2020), when the virus strain and national interventions were comparatively homogeneous. Exploratory analyses (Appendix ??) suggest treating R_0 as a global parameter. Each department $k = 1, \ldots, 94$ has local parameters $\mu_k = (I_k(0), R_k(0), \nu_k)$, with $\delta_k = R_0 \nu_k$, yielding a hierarchical structure with exchangeable locals and conditional independence across compartments.

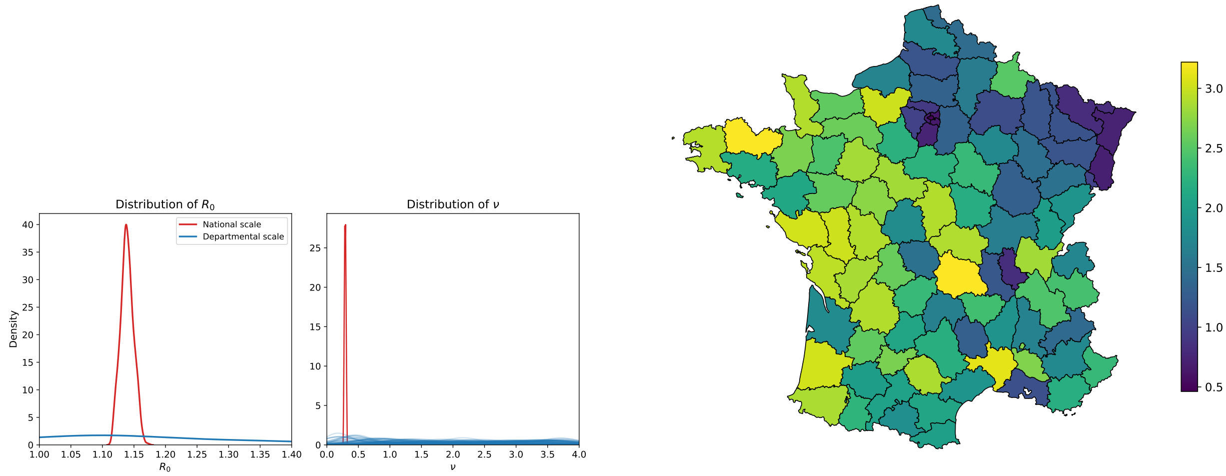
With $K = 94$ compartments and three local parameters per department plus R_0, the parameter space has dimension $d = 3 \times 94 + 1 = 283$. This high-dimensional setting is challenging for standard ABC; permABC-SMC exploits exchangeability and combines with the kernels of Section 2.6 to perform inference at this scale.

Figure 7a compares the posterior of R_0 from aggregated national data versus the 94 departmental time series: the departmental posterior is markedly more concentrated, reflecting the gain from jointly modelling spatial heterogeneity. The inferred local recovery rates are mapped in Figure 7b.

The SIR model is inevitably misspecified for real hospitalization data: the observation process is coarsely modelled and many mechanisms (reporting effects, delays, spatial interactions) are ignored. Some estimates of ν_k take implausibly large values, compensating for this misspecification. This application should therefore be viewed as a proof of scalability for permABC rather than a fully realistic epidemiological analysis. To confirm that the method itself is sound, Appendix ?? presents a synthetic-data experiment with known ground-truth parameters: permABC-SMC recovers the true R_0 and local ν_k accurately, together with well-calibrated posterior predictive envelopes, for three values of the noise parameter $\sigma \in \{0.01, 0.05, 0.10\}$.

4 Conclusion

We have introduced *permABC*, a novel framework for Approximate Bayesian Computation (ABC) tailored to hierarchical models with exchangeable structure. By exploiting the



(a) Posterior for R_0 and ν_k; departmental ($K = 94$, solid) vs. national ($K = 1$, dashed).

(b) Posterior mean ν_k by department.

Figure 7: SIR model on French COVID-19 data ($K = 94$ departments, model (13)).

invariance of compartments under permutations, permABC restores identifiability among local parameters and significantly improves acceptance rates in high-dimensional settings.

Extending this framework, we developed a sequential version, *permABC-SMC*, which adapts classical ABC-SMC algorithms to permutation-based acceptance rules. We further proposed two complementary strategies: *Over-Sampling*, which enriches the simulation space early on to facilitate acceptance, and *Under-Matching*, which relaxes the matching constraint to allow robust early-stage inference. These approaches provide new tools for navigating pseudo-posterior sequences, enabling both computational efficiency and resilience to model misspecification.

Our theoretical results show that permABC retains the desirable convergence properties of classical ABC algorithms in the small-tolerance regime. Furthermore, numerical experiments highlight the efficiency and stability of permABC in high-dimensional scenarios, as well as its robustness to model misspecification where coordinate-wise approaches like ABC-Gibbs may fail.

We also clarified the connection between permABC and Wasserstein ABC: at the level of compartment-level empirical measures, permABC is a natural hierarchical analogue of WABC, and in the one-dimensional case the two approaches coincide. Asymptotic arguments suggest a WABC-style concentration on the global parameter as $K \to \infty$, while the over-sampling limit $M \to \infty$ is a qualitatively different object acting as an algorithmic bridge rather than an inferential target. On the computational side, warm-started swap refinement reduces the amortised cost of the assignment step within SMC from $\mathcal{O}(K^3)$ to $\mathcal{O}(K^2)$ per particle.

Finally, we demonstrated the practical relevance of our method through an application to COVID-19 epidemic data, using a hierarchical SIR model across 94 departments. This application highlights the practical impact of the permABC: leveraging local heterogeneity leads to sharper inference on global parameters, opening new directions for scalable and robust likelihood-free inference.

All proposed methods are implemented in a dedicated Python package called `permabc`, available at `https://github.com/AntoineLuciano/permABC`. The repository also includes code to reproduce the figures and experiments presented in this paper.

Future directions include a deeper theoretical and empirical investigation of the Over-Sampling and Under-Matching strategies, with particular emphasis on the design of more efficient proposal distributions. An additional promising avenue is to explore joint or adaptive use of both strategies within a unified sequential framework: their complementarity may further enhance inference performance.

Disclosure Statement

The authors report there are no competing interests to declare.

Data Availability Statement

All code and data to reproduce the experiments are available at https://github.com/aluciano/permABC.

SUPPLEMENTARY MATERIAL

Supplementary proofs and details: Technical proofs, stratified permABC derivations, and additional experimental details. (PDF)

permABC Python package: Python package implementing permABC-SMC, Over-Sampling, and Under-Matching, together with all scripts to reproduce the figures in the paper. (GitHub repository)

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